NON-OBSERVATIONAL MODEL / ARCHIVIST EDITION

Lilium Aeternum:
A Model-Derived Four-Planet Candidate Architecture

Prepared for **Lily Chen**
Commissioned as a birthday dedication by **Xie Yao Zhong**
NOCTUA Celestial Research Lab - 21 July 2026

Figure 1. Artist's impression of Lilium Aeternum. This generated visual is not telescope imagery or observational evidence.

Registry: **NOR-LILY2026** System: **SYS-LC-2026** Designation: **NOCTUA-LILIUM-0721**

Abstract

This dossier defines a deterministic, model-derived four-planet architecture around an F8V stellar prior. The system is a private scientific visualisation created for the Lilium Aeternum commemorative archive; it is not an observed target, a confirmed exoplanet system or an official astronomical designation. A Fibonacci-sphere sampling method supplies a reproducible sky coordinate, while Keplerian scaling relates the adopted orbital periods to semi-major axes. Equilibrium temperatures, bulk compositions and astrobiology scores are illustrative scenario outputs constrained by broad planetary-physics priors rather than measured spectra or transit data.

Status and scope

The archive separates three categories of information: (1) deterministic model inputs, including the coordinate sample and reference epoch; (2) physically motivated estimates, including orbital periods and equilibrium temperatures; and (3) imaginative interpretation, including colours, symbolic meaning and possible biological analogues. No category should be represented as direct observation.

Key archive facts

Private name	Lilium Aeternum
Recipient	Lily Chen
Giver	Xie Yao Zhong
Archive edition	Archivist / US$500 Gift Edition
Registry	NOR-LILY2026
Scientific model code	SYS-LC-2026 / NOCTUA-LILIUM-0721

1. Stellar prior and model position

The host is assigned an F8V prior with a mass of 1.11 solar masses, a radius of 1.18 solar radii, an effective temperature of 6,260 K and a luminosity of 1.72 solar luminosities. These values define a coherent but hypothetical main-sequence host and are not derived from photometry or spectroscopy.

Deterministic position model

Sample index 89 is placed on a Fibonacci sphere. With golden angle g = pi(3 - sqrt(5)), longitude is theta = (89g) mod 2pi. The vertical term uses z = 1 - 2 frac(89 x 0.61803398875). Declination is compressed by a factor of 0.68 to represent the adopted synthetic stellar-population prior. Right ascension is the longitude converted to sidereal hours. The resulting model position is RA 23h 52m 45.8s, Dec +55 degrees 40 minutes 18.1 seconds, at an adopted distance of 72.6 pc (236.8 light-years).

These coordinates make the visualisation reproducible. They are not connected to a catalogued source and must not be used to claim a telescope detection.

Stellar parameters

Parameter	Adopted value	Status
Spectral class	F8V	Model prior
Mass	1.11 M_sun	Model prior
Radius	1.18 R_sun	Model prior
Effective temperature	6,260 K	Model prior
Luminosity	1.72 L_sun	Model prior
Age	3.2 billion years	Illustrative estimate
Metallicity	+0.07 dex	Illustrative estimate

2. Planetary architecture

Orbital periods follow the two-body approximation P_years = sqrt(a_AU^3 / M_star), where stellar mass is expressed in solar units. Eccentricities are low to moderate scenario inputs. Mutual interactions, resonances and long-term N-body stability have not yet been integrated.

Body	Model type	Mass	Radius	a	Period	T_eq
b	Rose-lit mineral terrestrial	1.50 M_E	1.12 R_E	0.163 AU	22.8 d	720 K
c	Pearlescent ocean super-Earth	2.60 M_E	1.38 R_E	1.34 AU	537.4 d	280 K
d	Lavender ringed gas giant	156 M_E	9.70 R_E	5.39 AU	4,335 d	139 K
e	Crystalline ice world	5.80 M_E	1.82 R_E	13.2 AU	16,620 d	89 K

Equilibrium temperature

The first-pass temperature relation scales with stellar luminosity and orbital distance. It assumes efficient heat redistribution and a broad albedo prior. Atmospheres, clouds, rotation, tidal heating and greenhouse effects can move surface conditions far from the listed equilibrium temperature.

System narrative

Planet b represents intensity and formation; planet c carries the central ocean-world scenario; planet d provides a visually dominant ringed giant and potential moon system; planet e preserves the cold outer archive. This narrative is an artistic layer placed on top of the physical model, not an observational classification.

3. Temperate-world and life assessment

NOCTUA-LILIUM-0721 c receives the highest model astrobiology score (64/100). Its assigned equilibrium temperature, ocean fraction and atmospheric scenario permit a temperate interpretation. This score is a ranking tool, not a probability of life.

Factors supporting the scenario

The adopted orbit is consistent with a temperate irradiation regime around the stellar prior. A 2.6 Earth-mass planet could plausibly retain a substantial atmosphere, while ocean coverage and tidal forcing from a faint ring or moon population may provide chemical and thermal gradients.

Factors that remain unknown

There are no measured spectra, transit depths, masses, albedos, atmospheric abundances or stellar-activity time series. A dense carbon-dioxide atmosphere could produce a strong greenhouse state; high cloud reflectivity could cool the surface; atmospheric loss or an ocean beneath a high-pressure ice layer would lead to different habitability conclusions.

Speculative morphology

If persistent oceans and energy gradients existed, the least speculative biological analogue would be microbial marine ecology. More complex swimmers, reef-like colonies or amphibious organisms are creative visual possibilities only. The model does not predict fish-like people, intelligent beings or any specific organism, and no biosignature is claimed.

Habitability is not habitation. A physically temperate scenario is only one prerequisite among many, and current astronomy has not confirmed life beyond Earth.

4. Research roadmap and observability

A real candidate would require an independently detected stellar source followed by repeated observations. The archive therefore defines the following roadmap as a methodological guide rather than a promise of future detection.

Priority	Proposed study	Scientific purpose
1	Cross-match the model coordinate against public stellar catalogues	Determine whether any real source can be associated without manufacturing identity
2	Synthetic transit and radial-velocity injection tests	Estimate the signal amplitude required for each planet scenario
3	N-body stability integration	Test whether the four adopted orbits remain dynamically coherent
4	Atmospheric escape and climate sensitivity grid	Bound the range of plausible conditions for planet c
5	Independent observational follow-up	Only this stage could create evidence for an actual astronomical candidate

Viewing note

The model coordinate has high northern declination. In a real observing context, visibility would depend on the observer's latitude, horizon, local sidereal time, weather and the brightness of an associated source. Because this coordinate is synthetic, the website's live map should be treated as an educational ephemeris display rather than a finder chart.

Archive access

The live private system can be opened with registry NOR-LILY2026 at https://noctua-celestial-lab.yao1230.chatgpt.site/gift/lily-chen. Model positions are advanced from the reference epoch using the listed orbital periods.

5. Limitations, naming status and references

Limitations

The system is generated from model priors and does not derive from a telescope dataset. The coordinate, distance, star, planets, compositions, temperatures and biological interpretations are synthetic. Rendering choices exaggerate sizes, colours and separations for legibility. The generated astronomical artwork is an artist's impression.

Naming status

Lilium Aeternum is a private commemorative designation in the NOCTUA archive. It does not alter scientific catalogues, confer property rights or replace names approved through the International Astronomical Union. The registry is intended as a personal gift and a transparent educational visualisation.

References

1. NASA Science. Orbits and Kepler's Laws. https://science.nasa.gov/solar-system/orbits-and-keplers-laws/

2. Kopparapu, R. K., et al. (2013). Habitable Zones Around Main-Sequence Stars: New Estimates. The Astrophysical Journal, 765:131. https://doi.org/10.1088/0004-637X/765/2/131

3. NASA Exoplanet Archive. Confirmed Planet and candidate-data context. https://exoplanetarchive.ipac.caltech.edu/

4. International Astronomical Union. Frequently Asked Questions on astronomical naming. https://www.iau.org/IAU/Science/What-we-do/FAQs.aspx

5. NOCTUA Celestial Research Lab. Lilium Aeternum deterministic model specification, revision 2026-07-21.

Prepared with care for Lily Chen

This dossier accompanies a birthday gift from Xie Yao Zhong. Its scientific transparency is part of the gift: imagination is allowed to be beautiful without being presented as evidence.